security â€“ 2023â€‘2024 Landscape â€“ Research Report**
â€‘level decisionâ€‘makers, security strategists and academic audiences*

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1. Introduction

nce of geopolitical tension, rapid digital transformation, and the commercialisation of advanced attack tools has cr

andards and academic evidence** (see the bibliography at the end). The analysis is organised around three core

nabled social engineering and the rise of automated phishing.**
in and cloudâ€‘native attack surfaces expanding the attack surface.**
rd mandatory resilience â€“ zeroâ€‘trust, NIS2, and the nextâ€‘generation NIST CSF.**

gital Defense, IBM Xâ€‘Force, CrowdStrike, Mandiant), peerâ€‘reviewed research (IEEE, ACM CCS, USENIX), a

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2. Key Findings

â€‘Enabled Social Engineering Is Redefining the Phishing Threat

Evidence	Details

Iâ€‘generated phishing** as a top emerging vector, noting a 35â€¯% higher clickâ€‘through rate than classic phis
d phishing emails powered by GPTâ€‘4â€‘style language models; success rate increased from 12â€¯% to **â‰ˆ
g cause of breaches (â‰ˆâ€¯36â€¯% of incidents). The report highlights a *new subâ€‘category* â€“ â€œAIâ€‘as
Shows a **57â€¯% rise** in successful credentialâ€‘theft attempts that leveraged AIâ€‘generated malicious attac
ers demonstrates that traditional oneâ€‘off training loses effectiveness after **4â€¯weeks**, especially against AI

Implications

xtâ€‘aware, multilingual phishing content at scale, reducing the cost of campaign generation.
basic heuristics are insufficient; AIâ€‘based detection (e.g., transformerâ€‘based classifiers) must be deployed.
en awareness programmes, simulated AIâ€‘phishing drills, and â€œjustâ€‘inâ€‘timeâ€• reminders are now best p

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ain and Cloudâ€‘Native Attack Surfaces Are Accelerating Systemic Risk

Evidence	Details

attacks** (e.g., software component compromises, dependencyâ€‘pivoting) as the secondâ€‘most frequent high
es an attackâ€‘graph model for **Kubernetes misâ€‘configurations** and demonstrates how a single compromise
% increase** in â€œcontainerâ€‘escapeâ€• attempts and a **28â€¯% rise** in â€œserverless injectionâ€• incide
â€‘billâ€‘ofâ€‘materials (SBOM) gaps** are cited in 61â€¯% of supplyâ€‘chain incidents, emphasizing the need fo
Management** subâ€‘category (ID.SCâ€‘1) mandating continuous validation of thirdâ€‘party components, proven
rds a **19â€¯% yearâ€‘onâ€‘year rise** in cyberâ€‘incidents targeting cloudâ€‘native workloads in the UK public

Implications

ct assets; the â€œtrust boundaryâ€• now resides in the software supply chain and cloud orchestration layers.

nce**, automated provenance scanning, and runtime integrity monitoring (e.g., eBPFâ€‘based attestation).
chain risk assessments; the U.S. **Cybersecurity Executive Orderâ€¯2022** is being operationalised through the

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and Frameworks Are Moving Toward Mandatory Resilience and Zeroâ€‘Trust

Evidence	Details

on** annual economic loss by 2025 if current riskâ€‘mitigation trajectories continue; calls for â€œresilient digital e
risk, and Privacy** pillars; aligns with ISO/IECâ€¯27001/27002 controls, providing a unified crossâ€‘jurisdictional
controls for **cloud, mobile, AI/ML**, and **privacyâ€‘byâ€‘design**; mandatory for firms seeking certification in >
eporting within 24â€¯h**, cyberâ€‘risk governance, and **penalties up toâ€¯â‚¬10â€¯M** or 2â€¯% of global turn
strates a **governmentâ€‘led threatâ€‘intel sharing platform** that reduces detection latency by 30â€¯% for partic
rganisations that have **implemented Zeroâ€‘Trust architectures** experience **45â€¯% fewer lateralâ€‘moveme

Implications

mmendation to a **regulatory expectation** in the EU (NIS2) and is being codified in the upcoming NIST CSFâ€¯
access control**, and **privacyâ€‘preserving data processing** to satisfy both security and dataâ€‘protection ma
, reduction in Mean Time to Detect/Respond), as reflected in the U.S. Department of Commerceâ€™s 2024 post

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3. Conclusion

heral IT concern* but a systemic, economicâ€‘level threat demanding coordinated action across policy, technolog

tive attack surface and the most vulnerable. Continuous, AIâ€‘enhanced awareness training and adaptive email
eyond organisational borders. Provenance (SBOM), runtime attestation, and secure orchestration pipelines must
nchored by **Zeroâ€‘Trust**, **privacyâ€‘byâ€‘design**, and **mandatory incident reporting** (NIS2, NIST CSFâ€

dentâ€‘response), and people (continuous training)**â€”will be best positioned to reduce the likelihood, impact, a

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4. Sources

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